JEE MAIN / PREVIOUS YEAR PAPER

JEE Main 2026 6th April Morning Shift

Question Paper with Answer Key

75	ENGLISH	DARK
QUESTIONS	LANGUAGE	FORMAT

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Chemistry 25 questions

Q1. MCQ (Single Correct Answer)

An oxide of iron contains 69.9% iron, its empirical formula, is :

(Given : Molar mass of Fe and O are 56 and $16\ \mathrm{g\ mol^{-1}}$ respectively.)

(A) FeO (B) $\mathrm{Fe_2O_3}$ (C) $\mathrm{Fe_3O_4}$ (D) $\mathrm{FeO_3}$

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Q2. MCQ (Single Correct Answer)

If shortest wavelength of hydrogen atom in Lyman series is x, then longest wavelength in Balmer series of $\mathrm{He^+}$is:

(A) $\frac{9x}{5}$ (B) $\frac{36x}{5}$ (C) $\frac{x}{4}$ (D) $\frac{5x}{9}$

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Q3. MCQ (Single Correct Answer)

Match the LIST-I with LIST-II

List-I Orbital		List-II Radial nodes and nodal plane	
A.	2s	I.	1 Radial node + two nodal planes
B.	3s	II.	1 Radial node + one nodal plane
C.	3p	III.	2 Radial nodes + No nodal plane
D.	4d	IV.	1 Radial node + No nodal plane

Choose the correct answer from the options given below:

(A) A-IV, B-I, C-III, D-II (B) A-IV, B-II, C-III, D-I

(C) A-III, B-I, C-IV, D-II (D) A-IV, B-III, C-II, D-I

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Q4. MCQ (Single Correct Answer)

The pairs among

$\mathrm{A} = \left[\mathrm{SO_3^{2-}}, \mathrm{CO_3^{2-}}\right], \mathrm{B} = \left[\mathrm{O_2^{2-}}, \mathrm{F_2}\right], \mathrm{C} = \left[\mathrm{CN^-}, \mathrm{CO}\right], \mathrm{D} = \left[\mathrm{NH_3}, \mathrm{H_3O^+}\right]$ and $\mathrm{E} = \left[\mathrm{MnO_4^{2-}}, \mathrm{CrO_4^{2-}}\right]$ that do not have similar Lewis dot structure are

(A) A, B and E (B) A and E (C) B, C and D (D) C and D

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Q5. MCQ (Single Correct Answer)

Arrange the following isothermal processes in order of the magnitude of the work $(\mathrm{p}-\mathrm{V})$ involved between states 1 and 2.

A. Expansion in single stage $\mathrm{w_A}$

B. Expansion in multi stages w_B

C. Compression in single stage $\mathrm{w_C}$

D. Compression in multi stages w_D

Choose the correct option.

(A) $|w_B| > |w_A| > |w_C| > |w_D|$

(B) $|w_C| > |w_D| > |w_A| > |w_B|$

(C) $|w_C| > |w_D| > |w_B| > |w_A|$

(D) $|w_B| > |w_A| > |w_D| > |w_C|$

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Q6. MCQ (Single Correct Answer)

When 0.25 moles of a non-volatile, non-ionizable solute was dissolved in 1 mole of a solvent the vapor pressure of solution was $x\%$ of vapor pressure of pure solvent. What is $x\%$?

(A) 50%

(B) 60%

(C) 70%

(D) 80%

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Q7. MCQ (Single Correct Answer)

One mole each of He and $\mathrm{A(g)}$ are taken in a 10 L closed flask and heated to 400 K to establish the following equilibrium.

$\mathrm{A(g) \rightleftharpoons B(g)}$

$\mathrm{K_c}$ for this reaction at 400 K is 4.0 . The partial pressures (in atm) of He and B(g) are respectively (at equilibrium)

(Assume $\mathrm{He, A(g)}$ and $\mathrm{B(g)}$ behave as ideal gases)

(Given : $\mathrm{R = 0.082\ L\ atm\ K^{-1}\ mol^{-1}}$)

(A) 3.28, 2.624

(B) 2.624, 3.28

(C) 3.28, 0.656

(D) 0.656, 6.56

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Q8. MCQ (Single Correct Answer)

Consider the following data.

Electrolyte	$\wedge_{\mathrm{m}}^{\circ}\left(\mathrm{S\ cm^2\ mol^{-1}}\right)$
$\mathrm{BaCl_2}$	x_1
$\mathrm{H_2SO_4}$	x_2
HCl	x_3

$\mathrm{BaSO_4}$ is sparingly soluble in water. If the conductivity of the saturated $\mathrm{BaSO_4}$ solution is $x\ \mathrm{S\ cm^{-1}}$ then the solubility product of $\mathrm{BaSO_4}$ can be given as (Here $\wedge_{\mathrm{m}} = \wedge_{\mathrm{m}}^{\circ}$)

(A) $\frac{10^6 x^2}{\alpha^2(x_1+x_2-2x_3)^2}$

(B) $\frac{x^2}{(x_1+x_2-2x_3)^2}$

(C) $\frac{\alpha^2(x_1+x_2-2x_3)^2}{10^6 x^2}$

(D) $\frac{x^2}{(x_1+x_2+2x_3)^2}$

Q9. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : Aluminium is more electropositive than thallium as the standard electrode potential value of $E^{\circ} Al^{3+}/Al$ is negative and $E^{\circ} Tl^{3+}/Tl$ is positive.

Statement II : The sum of first three ionization enthalpies of boron is very high when compared to that of aluminium. Due to this reason boron forms covalent compounds only and aluminium forms Al^{3+} ion.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q10. MCQ (Single Correct Answer)

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The correct statements among the following are :

A. Basic vanadium oxide is used in the manufacture of H_2SO_4.

B. The spin-only magnetic moment value of the transition metal halide employed in Ziegler-Natta polymerization is 2.84 BM .

C. The p-block metal compound employed in Ziegler-Natta polymerization has the metal in +3 oxidation state.

D. The number of electrons present in the outer most ' d ' orbital of metal halide employed in Wacker process is 8.

Choose the correct answer from the options given below :

(A) A and B Only (B) A, C and D Only (C) C and D Only (D) B, C and D Only

Q11. MCQ (Single Correct Answer)

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Match the LIST-I with LIST-II

List-I **Electronic configuration of tetrahedral metal ion**		**List-II** **Crystal Field Stabilization Energy (Δ_t)**	
A.	d^2	I.	-0.6
B.	d^4	II.	-0.8
C.	d^6	III.	-1.2
D.	d^8	IV.	-0.4

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-II, D-I

(B) A-III, B-I, C-IV, D-II

(C) A-III, B-IV, C-I, D-II

(D) A-II, B-I, C-IV, D-III

Q12. MCQ (Single Correct Answer)

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Which of the following are true about the energy of the given d-orbitals of a tetrahedral complex?

A. $\mathrm{d}_{xy} = \mathrm{d}_{xz} > \mathrm{d}_{x^2-y^2}$

B. $\mathrm{d}_{xy} = \mathrm{d}_{yz} > \mathrm{d}_{z^2}$

C. $\mathrm{d}_{x^2-y^2} > \mathrm{d}_{z^2} > \mathrm{d}_{xz}$

D. $\mathrm{d}_{x^2-y^2} = \mathrm{d}_{z^2} < \mathrm{d}_{xz}$

Choose the correct answer from the given below :

(A) A, B and D only (B) A and B only (C) B and D only (D) B, C and D only

Q13. MCQ (Single Correct Answer)

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$\mathrm{R_f}$ value for 2-methylpropene in a solvent system (Ethyl acetate + ether) is 0.42 . 2-methylpropene is treated with dilute $\mathrm{H_2SO_4}$ to give major organic product (X). R_f value for (X) in the same solvent system under identical condition will be:

(A) 0.42 (B) 0.82 (C) 0.62 (D) 0.12

Q14. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : 2, 6-diethylcyclohexanone and 6-methyl-2-n-propylcyclohexanone are metamers.

Statement II : 2, 2, 6, 6 - tetramethylcyclohexanone exhibits keto-enol tautomerism.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q15. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : Methane can be prepared by decarboxylation of sodium ethanoate, Kolbe's electrolysis of sodium acetate and reaction of $\mathrm{CH_3MgBr}$ with water.

Statement II : Methane cannot be prepared from unsaturated hydrocarbons and by Wurtz reaction.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q16. MCQ (Single Correct Answer)

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Given below are two statements :

Statement I : 3-phenylpropene reacts with HBr and gives secondary alkyl bromide having a chiral carbon atom as the major product.

Statement II : Aryl chlorides and aryl cyanides can be prepared by Sandmeyer reaction as well as Gattermann reaction.

In the light of the above statements, choose the correct answer from the options given below

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

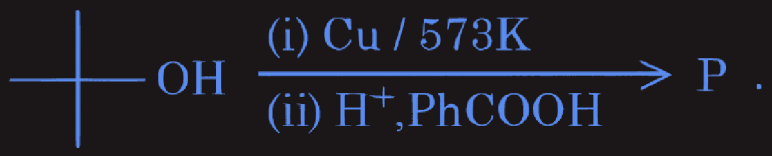
(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q17. MCQ (Single Correct Answer)

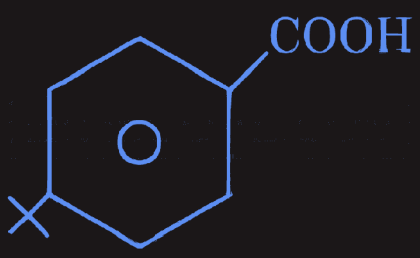
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Consider the following sequence of reactions

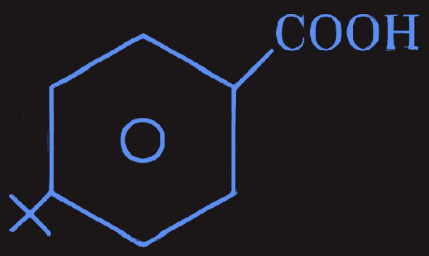


The major product P is :

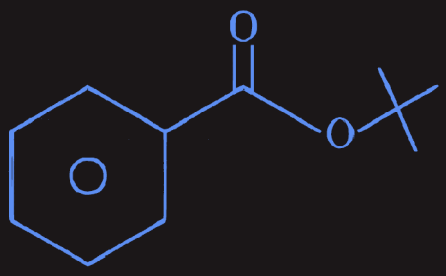
(A)



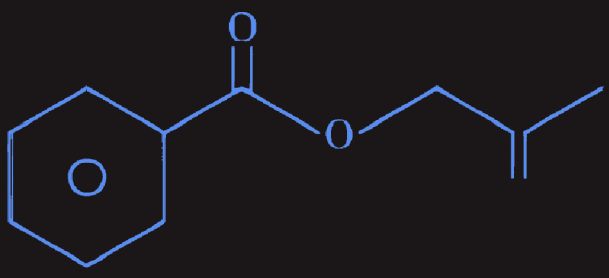
(B)



(C)



(D)



Q18. MCQ (Single Correct Answer)

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Arrange the following compounds according to increasing order of boiling points.

$n-C_4H_9OH(A), n-C_4H_9NH_2(B), n-C_4H_{10}(C)$ and $C_2H_5NHC_2H_5(D)$.

(A) $C < B < A < D$

(B) $D < C < B < A$

(C) $C < D < B < A$

(D) $D < B < A < C$

Q19. MCQ (Single Correct Answer)

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Match the List-I with List-II

List-I Deficiency Disease		List-II Vitamin	
A.	Scurvy	I.	Pyridoxine
B.	Convulsions	II.	Vitamin A
C.	Cheilosis	III.	Ascorbic Acid
D.	Xerophthalmia	IV.	Riboflavin

Choose the correct answer from the options given below :

(A) A-I, B-III, C-II, D-IV

(B) A-I, B-III, C-IV, D-II

(C) A-III, B-I, C-IV, D-II

(D) A-III, B-I, C-II, D-IV

Q20. MCQ (Single Correct Answer)

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Match the List-I with List-II

List-I Amino acid		List-II Positive reaction/Test for functional group present in side chain of amino acid	
A.	Glutamine	I.	Hinsberg's test
B.	Lysine	II.	Neutral $FeCl_3$ test
C.	Tyrosine	III.	Ceric ammonium nitrate test
D.	Serine	IV.	Hoffman bromamide degradation

Choose the correct answer from the options given below :

(A) A-IV, B-II, C-I, D-III

(B) A-IV, B-I, C-II, D-III

(C) A-III, B-II, C-I, D-IV

(D) A-IV, B-I, C-III, D-II

Q21. Numerical

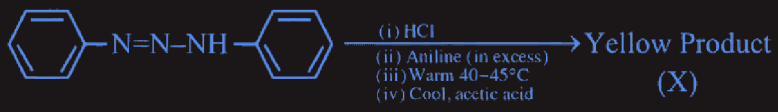
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First and second ionization enthalpies of lithium are $520\ \text{kJ mol}^{-1}$ and $7297\ \text{kJ mol}^{-1}$ respectively. Energy required to convert 3.5 mg lithium (g) into $Li^{2+}(g)$ $\left[Li(g) \rightarrow Li^{2+}(g)\right]$ is ____ kJmol^{-1}. (nearest integer)

[Molar mass of $Li = 7\ \text{g mol}^{-1}$]

Q22. Numerical

Consider the following sequence of reactions.



The percentage of nitrogen in the yellow product (X) formed is ____ $\%$.

(Nearest Integer)

(Given Molar mass in $\mathrm{gmol}^{-1} \mathrm{H}: 1, \mathrm{C}: 12, \mathrm{N}: 14$)

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Q23. Numerical

4.7 g of phenol is heated with Zn to give product X . If this reaction goes to 60% completion then the number of moles of compound X formed will be

____ $\times 10^{-2}$.(Nearest Integer)

(Given molar mass in $\mathrm{gmol}^{-1}: \mathrm{H}: 1, \mathrm{C}: 12, \mathrm{O}: 16$)

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Q24. Numerical

Sucrose hydrolyses in acidic medium into glucose and fructose by first order rate law with $t_{1/2} = 3$ hour. The percentage of sucrose remaining after 6 hours is

____ . (Nearest integer)

(Given : $\log 2 = 0.3010$ and $\log 3 = 0.4771$)

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Q25. Numerical

Consider the reaction $\mathrm{X} \rightleftharpoons \mathrm{Y}$ at 300 K . If $\Delta \mathrm{H}^{\theta}$ and K are $28.40\ \mathrm{kJ\ mol}^{-1}$ and 1.8×10^{-7} at the same temperature, then the magnitude of $\Delta \mathrm{S}^{\theta}$ for the reaction in $\mathrm{JK}^{-1}\ \mathrm{mol}^{-1}$ is ____ . (Nearest integer)

(Given : $\mathrm{R} = 8.3\ \mathrm{J\ K}^{-1}\ \mathrm{mol}^{-1}, \ln 10 = 2.3, \log 3 = 0.48, \log 2 = 0.30$)

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Mathematics 25 questions

Q26. MCQ (Single Correct Answer)

Let [•] denote the greatest integer function. If the domain of the function $f(x) = \sin^{-1}\left(\frac{x+[x]}{3}\right)$ is $[\alpha, \beta)$, then $\alpha^2 + \beta^2$ is equal to:

(A) 2 (B) 5 (C) 10 (D) 13

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Q27. MCQ (Single Correct Answer)

Let one root of the quadratic equation in x :

$\left(k^2-15k+27\right)x^2+9(k-1)x+18=0$

be twice the other. Then the length of the latus rectum of the parabola $y^2=6kx$ is equal to:

(A) 4 (B) 6 (C) 8 (D) 12

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Q28. MCQ (Single Correct Answer)

Let e_1 and e_2 be two distinct roots of the equation $x^2-ax+2=0$. Let the sets $\{a \in \mathbb{R} : e_1$ and e_2 are the eccentricities of hyperbolas $\} = (\alpha, \beta)$, and $\{a \in \mathbb{R} : e_1$ and e_2 are the eccentricities of an ellipse and a hyperbola, respectively $\} = (\gamma, \infty)$.

Then $\alpha^2+\beta^2+\gamma^2$ is equal to:

(A) 18 (B) 22 (C) 26 (D) 34

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Q29. MCQ (Single Correct Answer)

Let the set of all values of $k \in \mathbb{R}$ such that the equation $z(\bar{z}+2+i)+k(2+3i)=0, z \in \mathrm{C}$, has at least one solution, be the interval $[\alpha, \beta]$. Then $9(\alpha+\beta)$ is equal to:

(A) -10 (B) -8 (C) $10\sqrt{13}$ (D) $8\sqrt{13}$

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Q30. MCQ (Single Correct Answer)

The value of $1^3-2^3+3^3-\ldots+15^3$ is:

(A) 1706 (B) 1856 (C) 1982 (D) 2403

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Q31. MCQ (Single Correct Answer)

The sum of the first ten terms of an A.P. is 160 and the sum of the first two terms of a G.P. is 8 . If the first term of the A.P. is equal to the common ratio of the G.P. and the first term of the G.P. is equal to common difference of the A.P., then the sum of all possible values of the first term of the G.P. is:

(A) $\frac{34}{9}$ (B) $\frac{34}{13}$ (C) $\frac{32}{9}$ (D) $\frac{32}{13}$

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Q32. MCQ (Single Correct Answer)

The number of 4-letter words, with or without meaning, each consisting of two vowels and two consonants that can be formed from the letters of the word INCONSEQUENTIAL, without repeating any letter, is:

(A) 2670 (B) 2840 (C) 2920 (D) 3600

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Q33. MCQ (Single Correct Answer)

If the coefficients of the middle terms in the binomial expansions of $(1+\alpha x)^{26}$ and $(1-\alpha x)^{28}$, $\alpha \neq 0$, are equal, then the value of α is:

(A) 1 (B) $\frac{14}{13}$ (C) $\frac{27}{7}$ (D) $\frac{7}{27}$

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Q34. MCQ (Single Correct Answer)

A data consists of 20 observations $x_1, x_2, \ldots, x_{20}$. If $\sum_{i=1}^{20}(x_i+5)^2 = 2500$ and $\sum_{i=1}^{20}(x_i-5)^2 = 100$, then the ratio of mean to standard deviation of this data is :

(A) 2 : 1 (B) 3 : 1 (C) 3 : 2 (D) 4 : 1

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Q35. MCQ (Single Correct Answer)

A bag contains $(N+1)$ coins -N fair coins, and one coin with 'Head' on both sides. A coin is selected at random and tossed. If the probability of getting 'Head' is $\frac{9}{16}$, then N is equal to:

(A) 5 (B) 7 (C) 8 (D) 9

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Q36. MCQ (Single Correct Answer)

If the eccentricity e of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, passing through $(6, 4\sqrt{3})$, satisfies $15\left(e^2+1\right) = 34e$, then the length of the latus rectum of the hyperbola $\frac{x^2}{b^2} - \frac{y^2}{2(a^2+1)} = 1$ is:

(A) 10 (B) 20 (C) 25 (D) 30

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Q37. MCQ (Single Correct Answer)

Let chord PQ of length $3\sqrt{13}$ of the parabola $y^2 = 12x$ be such that the ordinates of points P and Q are in the ratio 1:2. If the chord PQ subtends an angle α at the focus of the parabola, then $\sin\alpha$ is equal to :

(A) $\frac{3}{5}$ (B) $\frac{4}{5}$ (C) $\frac{5}{13}$ (D) $\frac{12}{13}$

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Q38. MCQ (Single Correct Answer)

Let $0 < \alpha < 1, \beta = \frac{1}{3\alpha}$ and $\tan^{-1}(1-\alpha) + \tan^{-1}(1-\beta) = \frac{\pi}{4}$. Then $6(\alpha+\beta)$ is equal to:

(A) 6 (B) 7 (C) 8 (D) 9

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Q39. MCQ (Single Correct Answer)

Let $S = \{\theta \in (-2\pi, 2\pi) : \cos\theta + 1 = \sqrt{3}\sin\theta\}$.

Then $\sum_{\theta\in S}\theta$ is equal to :

(A) $-\frac{2\pi}{3}$ (B) $-\frac{4\pi}{3}$ (C) $\frac{2\pi}{3}$ (D) $\frac{4\pi}{3}$

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Q40. MCQ (Single Correct Answer)

Let the image of the point $\mathrm{P}(1,6,a)$ in the line $\mathrm{L}: \frac{x}{1}=\frac{y-1}{2}=\frac{z-a+1}{b}, b>0$, be $\left(\frac{a}{3}, 0, a+c\right)$. If $\mathrm{S}(\alpha, \beta, \gamma), \alpha>0$, is the point on L such that the distance of S from the foot of perpendicular from the point P on L is $2\sqrt{14}$, then $\alpha+\beta+\gamma$ is equal to:

(A) 19 (B) 20 (C) 21 (D) 22

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Q41. MCQ (Single Correct Answer)

Let a line L be perpendicular to both the lines $\mathrm{L}_1: \frac{x+1}{3}=\frac{y+3}{5}=\frac{z+5}{7}$ and $\mathrm{L}_2: \frac{x-2}{1}=\frac{y-4}{4}=\frac{z-6}{7}$.

If θ is the acute angle between the lines L and $\mathrm{L}_3: \frac{x-\frac{8}{7}}{2}=\frac{y-\frac{4}{7}}{1}=\frac{z}{2}$, then $\tan\theta$ is equal to:

(A) $\frac{3}{2}\sqrt{2}$ (B) $\frac{5}{2}\sqrt{2}$ (C) $\frac{5}{3}\sqrt{2}$ (D) $\frac{4}{3}\sqrt{2}$

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Q42. MCQ (Single Correct Answer)

The value of $\lim\limits_{x\to 0}\left(\frac{x^2\sin^2 x}{x^2-\sin^2 x}\right)$ is :

(A) 2 (B) 3 (C) 4 (D) 6

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Q43. MCQ (Single Correct Answer)

The value of the integral $\int\limits_{-\frac{\pi}{4}}^{\frac{\pi}{4}}\left(\frac{32\cos^4 x}{1+e^{\sin x}}\right)dx$ is :

(A) $4\pi+2$ (B) $3\pi+8$ (C) $3\pi+4$ (D) $4\pi+3$

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Q44. MCQ (Single Correct Answer)

The area of the region $\{(x,y): 0\le y\le 6-x, y^2\ge 4x-3, x\ge 0\}$ is :

(A) 8 (B) 9 (C) 12 (D) 15

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Q45. MCQ (Single Correct Answer)

Let e be the base of natural logarithm and let $f:\{1,2,3,4\}\to\{1,e,e^2,e^3\}$ and $g:\{1,e,e^2,e^3\}\to\{1,\frac{1}{2},\frac{1}{3},\frac{1}{4}\}$ be two bijective functions such that f is strictly decreasing and g is strictly increasing. If $\phi(x)=\left[f^{-1}\left\{g^{-1}\left(\frac{1}{2}\right)\right\}\right]^x$, then the area of the region $\mathrm{R}=\{(x,y): x^2\le y\le\phi(x), 0\le x\le 1\}$ is :

(A) $\frac{3-\log_e(2)}{3\log_e(2)}$ (B) $\frac{1}{3\log_e(2)}$

(C) $3+\log_e(2)$ (D) $\frac{3+\log_e(2)}{2+\log_e(3)}$

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Q46. Numerical

Let $A = \begin{bmatrix} -1 & 1 & -1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$ satisfy

$A^2 + \alpha(\text{adj}(\text{adj}(A))) + \beta(\text{adj}(A)(\text{adj}(\text{adj}(A)))) = \begin{bmatrix} 2 & -2 & 2 \\ -2 & 0 & -1 \\ 0 & 0 & -1 \end{bmatrix}$ for some $\alpha, \beta \in \mathbb{R}$.

Then $(\alpha - \beta)^2$ is equal to ____

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Q47. Numerical

Let the centre of the circle $x^2 + y^2 + 2gx + 2fy + 25 = 0$ be in the first quadrant and lie on the line $2x - y = 4$. Let the area of an equilateral triangle inscribed in the circle be $27\sqrt{3}$. Then the square of the length of the chord of the circle on the line $x = 1$ is ____ .

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Q48. Numerical

If $\vec{a} = \hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{j} - \hat{k}$ and $\vec{c}$ be three vectors such that $\vec{a} \times \vec{c} = \vec{b}$ and $\vec{a} \cdot \vec{c} = 3$, then $\vec{c} \cdot (\vec{a} - 2\vec{b})$ is equal to ____ .

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Q49. Numerical

For the functions $f(\theta) = \alpha \tan^2\theta + \beta \cot^2\theta$, and $g(\theta) = \alpha \sin^2\theta + \beta \cos^2\theta, \alpha > \beta > 0$, let $\min\limits_{0<\theta<\frac{\pi}{2}} f(\theta) = \max\limits_{0<\theta<\pi} g(\theta)$. If the first term of a G.P. is $\left(\frac{\alpha}{2\beta}\right)$, its common ratio is $\left(\frac{2\beta}{\alpha}\right)$ and the sum of its first 10 terms is $\frac{m}{n}, \gcd(m, n) = 1$, then $m + n$ is equal to ____ .

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Q50. Numerical

Let $y = y(x)$ be the solution of the differential equation $\left(x^2 - x\sqrt{x^2-1}\right)dy + \left(y\left(x - \sqrt{x^2-1}\right) - x\right)dx = 0, x \geq 1$. If $y(1) = 1$, then the greatest integer less than $y(\sqrt{5})$ is ____ .

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Physics 25 questions

Q51. MCQ (Single Correct Answer)

The density ρ of a uniform cylinder is determined by measuring its mass m, length l and diameter d. The measured values of m, l and d are 97.42 ± 0.02 g, 8.35 ± 0.05 mm and 20.20 ± 0.02 mm, respectively. Calculated percentage fractional error in ρ is ____ .

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(A) 0.63% (B) 0.82% (C) 0.72% (D) 0.25%

Q52. MCQ (Single Correct Answer)

The potential energy of a particle changes with distance x from a fixed origin as $V = \frac{A\sqrt{x}}{x+B}$, where A and B are constants with appropriate dimensions. The dimensions of AB are ____

(A) $\left[\mathrm{M}^1\,\mathrm{L}^{5/2}\,\mathrm{T}^{-2}\right]$

(B) $\left[\mathrm{M}^{3/2}\,\mathrm{L}^{5/2}\,\mathrm{T}^{-2}\right]$

(C) $\left[\mathrm{M}^1\,\mathrm{L}^2\,\mathrm{T}^{-2}\right]$

(D) $\left[\mathrm{M}^1\,\mathrm{L}^{7/2}\,\mathrm{T}^{-2}\right]$

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Q53. MCQ (Single Correct Answer)

The rain drop of mass 1 g , starts with zero velocity from a height of 1 km . It hits the ground with a speed of $5\ \mathrm{m/s}$. The work done by the unknown resistive force is ____ J.

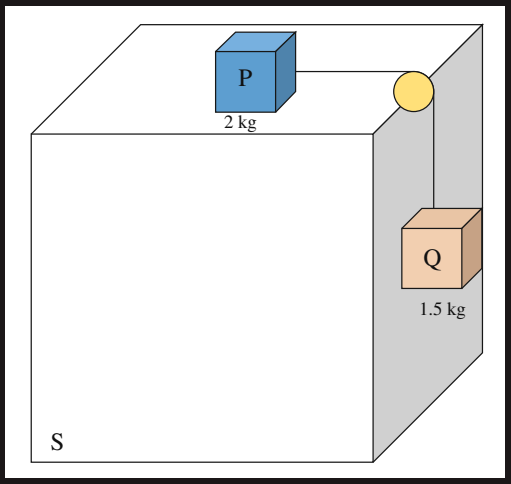
(take $\mathrm{g} = 10\ \mathrm{m/s^2}$)

(A) -8.75 (B) -8.35 (C) -9.55 (D) -9.98

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Q54. MCQ (Single Correct Answer)

Two blocks (P and Q) with respectively masses 2 kg and 1.5 kg are joined by a massless thread. These blocks are mounted on a frictionless pulley which is fixed on the edge of a cube (S), as shown in the figure below. Block P is positioned on the top surface which has no friction and block Q is in contact with side-surface, having coefficient friction μ. The cube (S) moves towards the right with acceleration of $\frac{g}{2}$, where g is gravitational acceleration. During this movement the block P and Q remain stationary. The value of μ is ____ (take $\mathrm{g} = 10\ \mathrm{m/s^2}$)



(A) 0.33 (B) 0.67 (C) 1 (D) 0.5

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Q55. MCQ (Single Correct Answer)

A lift of mass 1600 kg is supported by thick iron wire. If the maximum stress which the wire can withstand is $4 \times 10^8\ \mathrm{N/m^2}$ and its radius is 4 mm , then maximum acceleration the lift can take is ____ $\mathrm{m/s^2}$.

(take $\mathrm{g} = 10\ \mathrm{m/s^2}$ and $\pi = 3.14$)

(A) 2.56 (B) 3.89 (C) 4.32 (D) 5.16

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Q56. MCQ (Single Correct Answer)

A solid sphere of radius 4 cm and mass 5 kg is rotating (rotation axis is passing through the centre of the sphere) with an angular velocity of 1200 rpm . It is brought to rest in 10 s by applying a constant torque. The torque applied and the number of rotations it made before it comes to rest are ____ and ____ respectively.

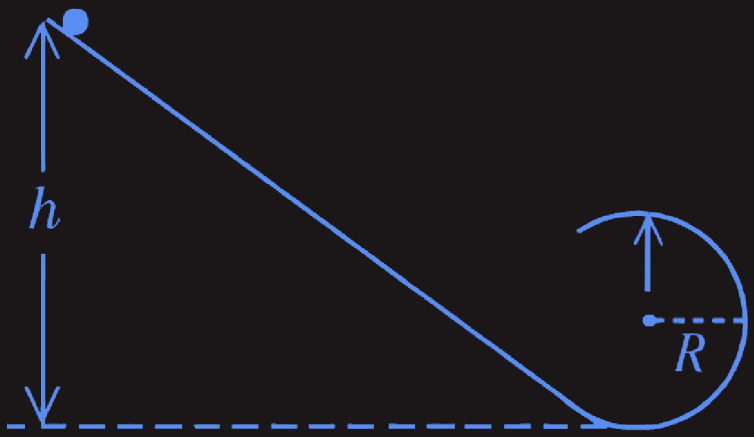
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(A) $0.128\pi\text{Nm}, 100$ (B) $0.0128\pi\text{Nm}, 50$ (C) $0.128\pi\text{Nm}, 50$ (D) $0.0128\pi\text{Nm}, 100$

Q57. MCQ (Single Correct Answer)

A smooth inclined plane ends in a vertical circular loop, as shown in the figure. A small body is released from height h as shown. If the body exerts a force of three times its weight on the plane at the highest point of circle then the height $h = \alpha R$. The value of α is ____

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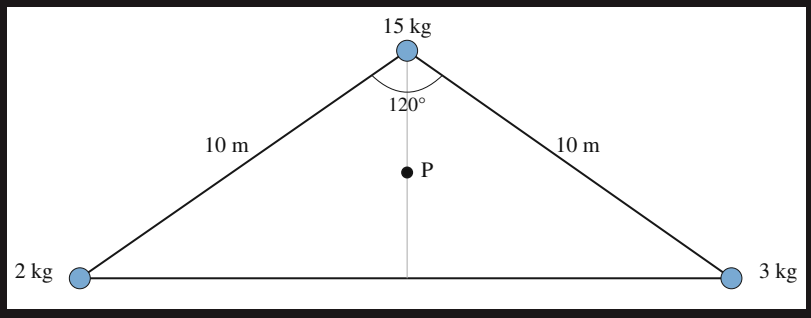


(A) 2 (B) 4 (C) 3 (D) 6

Q58. MCQ (Single Correct Answer)

The position of center of mass of three masses $2\text{ kg}, 3\text{ kg}$ and 15 kg placed with respect to mid point (p) of normal bisector, as shown in the figure is ____ .

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(A) $\left(\frac{\sqrt{3}}{4}, 1.25\right)$ (B) $\left(\frac{\sqrt{3}}{4}, 1.0\right)$

(C) $(0, 0)$ (D) $(1.25, 0)$

Q59. MCQ (Single Correct Answer)

The two wires A and B of equal cross-section but of different materials are joined together. The ratio of Young's modulus of wire A and wire B is 20/11. When the joined wire is kept under certain tension the elongations in the wires A and B are equal. If the length of wire A is 2.2 m , then the length of wire B is

____ m.

(A) 1.1 (B) 2.22 (C) 1.21 (D) 4.44

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Q60. MCQ (Single Correct Answer)

Two closed vessels of same volume are joined through a narrow tube and both vessels are filled with air of pressure 90 kPa and temperature 400 K . Keeping the temperature of one vessel constant at 400 K the second vessel temperature is raised to 500 K . The final pressure in the vessels is ____ kPa .

(A) 100 (B) 120 (C) 90 (D) 105

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Q61. MCQ (Single Correct Answer)

In interference experiment the path difference between two interfering waves at a point A on the screen is $\lambda/3$, where λ is the wavelength of these waves, and at another point B the path difference is $\lambda/6$. The ratio of intensities at points A and B is ____ .

(A) 3 (B) 4 (C) 1/3 (D) 1/4

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Q62. MCQ (Single Correct Answer)

A particle is executing simple harmonic motion. Its amplitude is A and time period is 5 sec . The time required by it to move from $x = A$ to $x = \frac{A}{\sqrt{2}}$ is ____ sec.

(A) 1/4 (B) 5/4 (C) 5/8 (D) 3/8

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Q63. MCQ (Single Correct Answer)

A thin half ring of radius 35 cm is uniformly charged with a total charge of Q coulomb. If the magnitude of the electric field at centre of the half ring is $100\ \mathrm{V/m}$, then the value of Q is ____ nC .

$\left(\epsilon_0 = 8.85 \times 10^{-12} \mathrm{C^2/Nm^2}\right.$ and $\left.\pi = 3.14\right)$

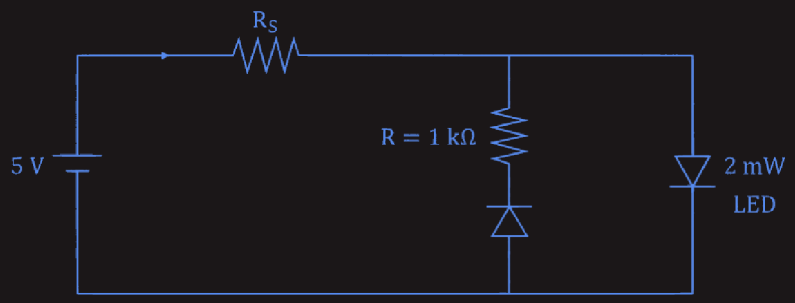
(A) 2.14 (B) 2.44 (C) 3.25 (D) 0.7

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Q64. MCQ (Single Correct Answer)

The maximum rated power of the LED is 2 mW and it is used in the circuit with input voltage of 5 V as shown in the figure below. The current through resistance R_S is 0.5 mA .

The minimum value of the resistance of R_S, to ensure that the LED is not damaged is ____ $\mathrm{k\Omega}$.



(A) 6 (B) 2 (C) 4 (D) 5

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Q65. MCQ (Single Correct Answer)

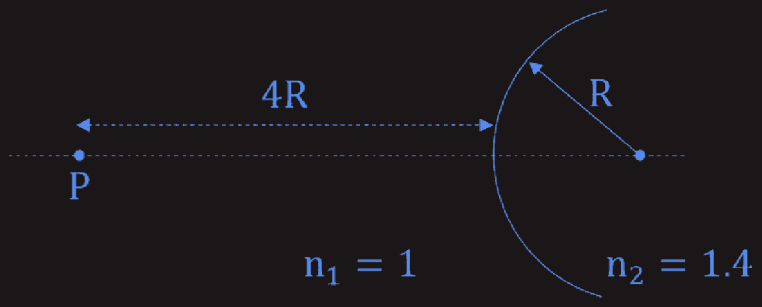
A point light source emits E.M. waves in free space. A detector, placed at a distance of L m, measures the intensity as I_o. The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45°. The measured intensity at new location will be ____ .

(A) $\frac{I_o}{4}$ (B) I_o (C) $\frac{I_0}{\sqrt{2}}$ (D) $\frac{I_o}{2}$

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Q66. MCQ (Single Correct Answer)

A spherical interface lens of radius R separates two media of refractive indices 1 and 1.4 respectively as shown in the figure below. A point source is placed at a distance of $4R$ in front of spherical interface. The magnitude of the magnification of point source image is ____ .



(A) 1.66 (B) 2.33 (C) 2.66 (D) 1.33

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Q67. MCQ (Single Correct Answer)

A small cube of side 1 mm is placed at the centre of a circular loop of radius 10 cm carrying a current of 2 A . The magnetic energy stored inside the cube is $\alpha \times 10^{-14}$ J. The value of α is ____ .

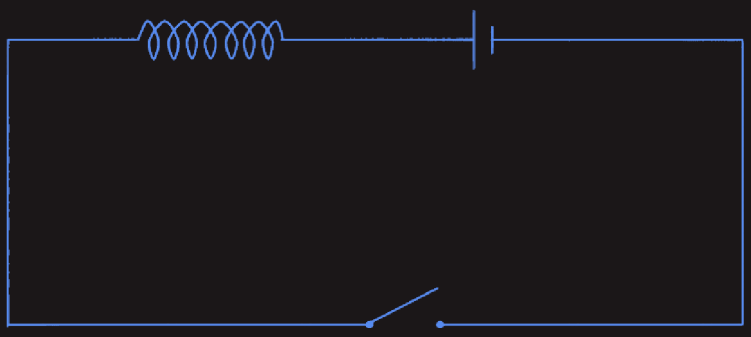
$\left(\mu_o = 4\pi \times 10^{-7}\mathrm{Tm/A}, \pi = 3.14\right)$

(A) 6.28 (B) 6.28×10^{-6} (C) 628 (D) 6.28×10^{-4}

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Q68. MCQ (Single Correct Answer)

An inductor of inductance 10 mH having resistance of 100Ω is connected to battery of E.M.F. 1.0 V through a switch as shown in the figure below. After switch is closed, the ratio of instantaneous voltages across the inductor when the current passing through it is 2 mA and 4 mA is ____ .



(A) 4/3 (B) 3/4 (C) 5/3 (D) 3/5

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Q69. MCQ (Single Correct Answer)

The ratio of momentum of the photons of the 1st and 2nd line of Balmer series of Hydrogen atoms is α/β. The possible values of α and β are:-

(A) 27 and 20 (B) 3 and 16 (C) 5 and 36 (D) 20 and 27

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Q70. MCQ (Single Correct Answer)

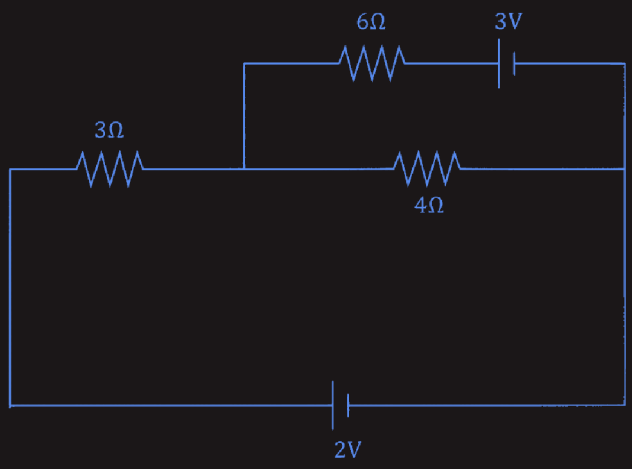
A LCR series circuit driven with $E_{rms} = 90\ \text{V}$ at frequency $f_{\text{d}} = 30\ \text{Hz}$ has resistance $R = 80\Omega$, an inductance with inductive reactance $X_L = 20.0\Omega$ and capacitance with capacitive reactance $X_C = 80.0\Omega$. The power factor of the circuit is ____ .

(A) 0.8 (B) 0.64 (C) 0.9 (D) 0.5

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Q71. Numerical

Refer to the circuit diagram given below. The heat generated across the 6Ω resistance in 100 second is $\frac{\alpha}{100}$ J. The value of α is ____ . (Nearest integer)



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Q72. Numerical

An unpolarized light of intensity I_0 passes through polarizer and then through a certain optically active solution and finally it goes to analyser. If the angle between analyser and polariser is 0° and intensity of light emerged from analyser is $\frac{3}{8}I_0$, the angle of rotation of the light by the solution with respect to analyser is ____ degrees.

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Q73. Numerical

The energy released when $\frac{7}{17.13}$ kg of $^{7}_{3}\mathrm{Li}$ is converted into $^{4}_{2}\mathrm{He}$ by proton bombardment is $\alpha \times 10^{32}\mathrm{eV}$. The value of α is ____ . (Nearest integer) (Mass of $^{7}_{3}\mathrm{Li} = 7.0183\mathrm{u}$, mass of $^{4}_{2}\mathrm{He} = 4.004\mathrm{u}$, mass of proton $= 1.008\mathrm{u}$ and $1\mathrm{u} = 931\mathrm{MeV/c^2}$ and Avogadro number $= 6.0 \times 10^{23}$)

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Q74. Numerical

A three coulomb charge moves from the point $(0, -2, -5)$ to the point $(5, 1, 2)$ in an electric field expressed as $\vec{E} = 2x\hat{\mathrm{i}} + 3\mathrm{y}^2\hat{\mathrm{j}} + 4\hat{\mathrm{k}}\mathrm{N/C}$. The work done in moving the charge is ____ J.

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Q75. Numerical

A certain gas is isothermally compressed to $\left(\frac{1}{3}\right)^{\mathrm{rd}}$ of its initial volume ($V_\circ = 3$ litre) by applying required pressure. If the bulk modulus of the gas is $3 \times 10^5\ \mathrm{N/m^2}$, the magnitude of work done on the gas is ____ J.

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Answer Key

Chemistry

1. **B**	2. **A**	3. **D**	4. **B**	5. **C**
6. **D**	7. **A**	8. **A**	9. **A**	10. **C**
11. **C**	12. **A**	13. **D**	14. **C**	15. **D**
16. **C**	17. **C**	18. **C**	19. **C**	20. **B**
21. **4**	22. **21**	23. **3**	24. **25**	25. **34**

Mathematics

26. **B**	27. **D**	28. **C**	29. **A**	30. **B**
31. **A**	32. **D**	33. **D**	34. **B**	35. **B**
36. **A**	37. **A**	38. **B**	39. **B**	40. **C**
41. **B**	42. **B**	43. **B**	44. **B**	45. **A**
46. **4**	47. **80**	48. **3**	49. **1279**	50. **3**

Physics

51. **B**	52. **D**	53. **D**	54. **B**	55. **A**
56. **D**	57. **B**	58. **A**	59. **C**	60. **A**
61. **C**	62. **C**	63. **A**	64. **B**	65. **B**
66. **A**	67. **A**	68. **A**	69. **D**	70. **A**
71. **3477**	72. **30**	73. **6**	74. **186**	75. **989**

For numerical answers given as a range, any value within the range is correct.

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